Lab 13: Darwin & Evolution

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this lab, you should be able to:

1. Use the **variation → overproduction → differential reproductive success** logic to predict trait change in a changing environment.
2. Use a **Punnett square** for one gene and track how **selection** changes **allele frequencies** in a single generation.
3. Apply the biological meaning of **fitness** to messy, realistic vignettes.
4. Choose which **line of evidence** tests which kind of claim about history and relatedness.
5. Explain why **individuals do not evolve** while **populations** can.

Overview

You are not rehearsing definitions today—you are **playing biologist**. Each part is a **mini scenario**: you will make a **prediction**, watch a **story outcome** (on paper), and then **defend** whether natural selection, evidence, or "common sense" actually fits. Part 3 adds a simple **genetic model** so you can connect **Mendelian ratios** to **allele frequency change** after selection. The thread is simple: evolution is **population-level**, **evidence-based**, and **mechanistic**, not a ladder of "better animals."

Materials Needed

- Pencil or pen
- This worksheet
- Calculator (optional, for division)

Part 1: Natural selection in two habitats (Fictional island stick insects)

Imagine an island with two habitats: open lava (dark rock) and mossy groves (bright green). A single species of stick insect comes in two heritable color morphs: charcoal and leaf-green. Insect-eating birds hunt by sight.

1. Before you read the "results," write a hypothesis.

Hypothesis 1 (lava plots): Over many generations on open lava, do you expect charcoal or leaf-green morphs to become more common? **Circle one:** Charcoal / Leaf-green / No change — then **one sentence why:**

Hypothesis 2 (moss groves): In bright green moss, do you expect charcoal or leaf-green to become more common? **Circle one:** Charcoal / Leaf-green / No change — **one sentence why:**

Model assumptions: These insects and numbers are fictional. Color is heritable; apart from bird predation, both morphs have equal opportunities to reproduce. A matching trend supports this model but does not exclude other causes.

2. "Field notes" (given): After 30 generations, lava plots are ~85% charcoal; moss groves are ~80% leaf-green. Bird stomach contents show they capture what stands out against the background.

Match mechanism to story: What is the **selective agent**? What is being **selected** (which phenotypes leave more offspring)?

- Selective agent:
- What increases in each habitat (one short phrase each):
- On lava:
- In moss:

3. Revision check: Did your hypotheses match the outcomes? If not, what environmental detail would you add to your model next time?

Part 2: What "fitness" means (short scenarios)

Evolutionary fitness is relative reproductive success within the same population and environment. Compare descendants that contribute to later generations, rather than body size or survival alone.

4. One-sentence definition of evolutionary fitness (your words):

5. Compare reproductive success. Within each same-species pair, write **higher** or **lower** based on the supplied lifetime data. Do not rank moose against salmon.

Competitor story	Fitness (higher / lower)
Bull moose A: huge antlers, wins fights, **0 calves**	
Bull moose B: smaller antlers, avoids fights, **3 calves later reproduce**	
Salmon A: dies after spawning; **20 offspring later reproduce**	
Salmon B: lives longer; **0 offspring reproduce**	

6. Puzzle: An antibiotic stops bacteria from dividing without killing them. Their untreated relatives continue dividing. Does survival alone establish **higher bacterial fitness** in evolutionary terms? Explain.

Part 3: Genetics of color — Punnett square and one generation on lava

We model the same species with one gene and two alleles, with complete dominance:

- **C** = dominant allele → **charcoal** phenotype (`CC` or `Cc`)
- **c** = recessive allele → **leaf-green** phenotype only when homozygous `cc`

Two heterozygotes mate: `Cc` × `Cc`.

7. Write the phenotype for each genotype.

Genotype	Phenotype (charcoal or leaf-green)
CC	
Cc	
cc	

8. Complete the Punnett square for `Cc` × `Cc`. Each parent makes gametes **C** and **c**.

	C	**c**
C		
c		

9. From the square, give the expected zygote ratios (use form 1 ___ : 2 ___ : 1 ___).

10. Expected *phenotype* ratio among those zygotes (charcoal vs leaf-green):

11. One generation on lava (differential survival). Imagine **400** new zygotes in the exact proportions above: **100** `CC`**, 200** `Cc`**, 100** `cc`. On open lava, bright green stands out to birds. Use these **relative survival** values (viability before reproduction):

- **w_CC** = 1.0; **w_Cc** = 1.0; **w_cc** = 0.25

Prediction: Before you calculate, do you expect the frequency of the `c` allele among survivors to **go up**, **go down**, or **stay the same**? One sentence why.

12. Count survivors (multiply number of each genotype by its `w`):

Genotype	Number before selection	× w	Survivors (individuals)
CC	100	× 1.0	
Cc	200	× 1.0	
cc	100	× 0.25	
Total	400	—	

13. Allele counting among survivors only. Each diploid individual carries **2** alleles.

- Total **C** alleles in survivors: *(Hint:* `CC` *contributes 2 per individual,* `Cc` *contributes 1.)*
- Total **c** alleles in survivors: *(Hint:* `Cc` *contributes 1 per individual,* `cc` *contributes 2.)*
- Total alleles = 2 × (number of survivors):

14. Frequencies after selection (among survivors):

- Frequency of **c** (call it q after selection): _____ *(= total c alleles ÷ total alleles)*
- Frequency of **C** (p after selection): _____

15. Compare to the starting zygote pool (before selection): there were **400 C** and **400 c** alleles in 400 individuals, so $p = q = 0.5$. Did q **increase** or **decrease** after one generation on lava? Does that match Part 1's lava story?

16. (Optional stretch) On **moss**, suppose leaf-green blends in best. Sketch a pattern of relative survival (`w`) across genotypes that would favor `cc` *without* writing a full calculation.

Part 4: Matching scientific claims to the strongest line of evidence

Evidence menu (pick one letter per row):
A Fossils & stratigraphy — order of forms in rock layers; transitional morphology in time.
B Homologous structures (comparative anatomy) — same underlying plan, different function (e.g., limb bones).
C Biogeography — where species live relative to geology and dispersal history.
D Molecular sequences (DNA/proteins) — similarity and branching when anatomy is misleading.

17. Which line of evidence best addresses each claim? Write **A**, **B**, **C**, or **D**.

Claim	Best evidence
"Whales look nothing like hippos today—yet detailed **skeletal homologies** (and fossils) link whales to even-toed ungulates." *(Which line is the primary match for unexpected anatomical correspondences?)*	
"Why are **marsupials** so concentrated in Australia compared with placental mammals on other continents?"	
"Can we see **fish-to-tetrapod** intermediates and their **order in the fossil record**?"	
"When **convergent evolution** makes two groups look similar, which data often resolve **true branching order**?"	

18. Convergence vs homology (bat vs bird). Bat wings and bird wings are both used for flight but evolved from different forelimb histories.

a) Compare two levels: are bat and bird **forelimb bones homologous**, and are their **powered-flight wings analogous**? Explain how shared ancestry and independent flight adaptations can both be true.

b) What would you compare next to decide **deep ancestry** vs **convergence** (name at least two: e.g., bone-by-bone forelimb, fossils, genes)?

19. Burrowing look-alikes. Two distantly related lineages evolve **similar-looking** cylindrical bodies for burrowing.

a) Is surface similarity alone strong evidence of **recent** common ancestry? **Yes / No** — why?

b) What evidence types from the A–D menu would you combine to test history?

Part 5: Individuals vs populations (inheritance and evolution)

You already know DNA replication and the universal code (Module 07). Now stress-test "population thinking."

20. Thought experiment — "Super-Soldier Serum" (fiction): A person's muscles triple in size after an injection, but **the treatment creates no heritable change in gametes**.

Prediction: Can their children inherit bigger muscles **because of the injection**? **Yes / No. Explain using "individual vs population" and heritability:**

21. Fix the broken claims:

Claim	True / False	Fix in one sentence
“That bird evolved a sharper beak during its life to crack nuts.”		
“Heritable deeper beaks became more common across generations because deep-beaked birds left more reproducing offspring during a drought.”		

22. Synthesis (3–4 sentences): How does the **near universality of the genetic code**, together with shared cellular features and DNA sequence similarities, support **common ancestry**? Why are several independent evidence types stronger than one similarity alone?